

F1 Strategy Engineer — Final Year Feature Plan

Curated Phase 2 feature set: uncertainty-aware, multi-strategy race optimization (final-year extension of the Phase 1 undercut/degradation engine)

This plan curates the strongest ideas from the initial Phase 2 brainstorm into six features that are each technically distinct from the Phase 1 system, realistically buildable on the existing FastF1 + XGBoost + Streamlit pipeline within a final-year timeline, and defensible as genuine research contributions rather than incremental tweaks. High-effort, high-uncertainty items from the original brainstorm (full multi-agent reinforcement learning, wet-weather modeling, LSTM/state-space degradation models) were deliberately excluded or downgraded to keep scope realistic.

Proposed Paper Title

"Uncertainty-Aware Monte Carlo Optimization of Formula 1 Pit Strategy with Empirically-Calibrated Competitor Reactions and Stochastic Race Events"

1. Probabilistic Tyre Degradation Model (Quantile Regression)

What it is: Instead of XGBoost predicting a single `lap_time_delta`, train it (or a wrapper such as NGBost, or XGBoost with quantile/pinball loss at $\tau = 0.05, 0.5, 0.95$) to output a distribution rather than a point estimate.

Why it matters: This is the single highest-leverage change. Every downstream feature (Monte Carlo simulation, risk-aware decisions, confidence scores) depends on having uncertainty bounds instead of one number. Reviewers will immediately recognize “point prediction → probabilistic prediction” as a real methodological shift, not a tweak.

Implementation approach: XGBoost supports `reg:quantileerror` (recent versions) or you can train 3 separate models at different quantiles. Output becomes (p5, p50, p95) per lap instead of one float.

Paper angle: “Does modeling degradation uncertainty change strategic recommendations versus point-estimate models?” — a directly comparable ablation study.

2. Multi-Strategy Monte Carlo Optimizer

What it is: Expand `simulation.py` to evaluate a full strategy space: Stay Out, Pit→Soft, Pit→Medium, Pit→Hard, Pit lap+1, Pit lap+2, and two-stop strategies — each run through $N \geq 1000$ Monte Carlo rollouts sampling from Feature 1's degradation distributions.

Why it matters: This is the “undercut detector → strategy optimizer” reframing, and it's the must-have change that redefines your research question entirely, replacing the binary Stay Out vs. Pit Now comparison.

Implementation approach: For each candidate strategy, sample degradation trajectories from the quantile model, sum simulated race time across rollouts, and obtain an empirical outcome distribution per strategy. Select the strategy that minimizes expected race time (or maximizes $P(\text{finish ahead of rival})$).

Paper angle: Example output: “Pit→Medium (lap 32): 68% $P(\text{net gain})$, expected gain $+2.3s \pm 1.1s$ ” — far richer than a binary OPEN/CLOSED signal.

3. Adaptive Decision Rule (Removing the Fixed 1.5s Threshold)

What it is: Replace the hardcoded rule with: Expected Net Gain = Fresh-Tyre Gain – Pit-Lane Loss – Traffic/Position Cost, evaluated continuously against the actual rival gap, remaining laps, and circuit-specific pit loss.

Why it matters: You already identified the fixed 1.5-second / 3-lap rule as an arbitrary heuristic in Phase 1. Fixing your own documented weakness gives a strong, easy-to-defend narrative for a committee: “we found this limitation, and here is the principled fix.”

Implementation approach: This falls out almost for free once Features 1 and 2 exist — it becomes a query against the Monte Carlo output rather than a comparison to a fixed constant.

4. Rival Reaction Model (Lightweight Two-Agent, Not Full Multi-Agent RL)

What it is: Rather than modeling a full multi-agent race, implement a tractable reactive model: when Driver A pits, Driver B has a calibrated probability of reacting by pitting on the following lap.

Why it matters: Delivers real strategic-interaction value without the scope and risk of full multi-agent reinforcement learning, which even the original brainstorm correctly flags as a stretch goal you should not start with.

Implementation approach: Mine historical FastF1 data for a “cover rate” — $P(\text{rival pits within 1 lap of an undercut attempt})$, conditioned on gap size. Feed this probability into the Monte Carlo simulator as a branching event.

Paper angle: “Incorporating empirically-calibrated competitor reaction probabilities into undercut strategy optimization.”

5. Safety Car / VSC as Stochastic Simulation Events

What it is: Inject Safety Car / VSC probability — estimated per circuit from historical race data (e.g. Baku or Singapore have much higher SC likelihood than Monza) — into the Monte Carlo rollouts as random events that collapse effective pit-loss cost.

Why it matters: Cheap to add given Feature 2's architecture, very intuitive to demo (“pitting under a Safety Car is nearly free”), and models genuinely realistic race strategy behavior that Phase 1 ignores entirely.

Implementation approach: Precompute a per-circuit SC rate table from the 2022–2025 data you already hold. Sample a Bernoulli event per rollout lap; if SC triggers, reduce `pit_loss_delta` by roughly 60–70%.

6. Explainability Layer + Historical Backtesting

What it is: Two components that work together as your evaluation methodology:

- SHAP on the degradation model, so any recommendation (e.g. “Pit Now”) shows top contributing factors

(tyre age, rival gap, degradation slope, pit loss) — turning a black-box number into decision support an engineer could trust.

- Backtesting against real historical races: replay 30–50 historical races lap-by-lap, feed real telemetry state into the Phase 2 optimizer at each decision point, and compare its recommended strategy against what the team actually did.

Why it matters: The original brainstorm correctly calls backtesting “critical”: without it, the paper’s only evidence is R^2 , which reviewers will rightly say proves the regression fits well but not that the strategy is better. This feature is your actual results/evaluation section.

Implementation approach: Report metrics such as simulated race-time delta, undercut success rate, and strategy regret across the backtested races.

Feature Summary

#	Feature	Type	Depends On
1	Probabilistic degradation model (quantile regression)	Foundational / Must-have	—
2	Multi-strategy Monte Carlo optimizer	Must-have	Feature 1
3	Adaptive decision rule (remove 1.5s threshold)	Must-have	Features 1, 2
4	Rival reaction model (2-agent, calibrated)	Strong addition	Feature 2
5	Safety Car / VSC stochastic events	Strong addition	Feature 2
6	Explainability (SHAP) + historical backtesting	Critical evaluation	Features 1–5

Note: Features excluded from the original brainstorm (full multi-agent reinforcement learning, wet-weather compound-switching strategy, and temporal/state-space degradation models) remain viable stretch goals if time permits, but are not required for a strong, defensible final-year paper.